

A common orientation sign for the radial derivative average

MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial

theorem

Identifies the whole minor vector using one signed Jacobian integral.

Statement

Let $m, N \in \mathbb{N}_0$, put $n = m + 1 > 0$, and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p, q : E \rightarrow \mathbb{R}$ be continuous real seminorms with specified positive constants a_p, b_p, a_q, b_q such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty \quad (x \in E).$$

Both p and q are norms. For each $s \in \{p, q\}$ define the closed unit ball, unit sphere, and real reference volume by

$$K_s = \{x : s(x) \leq 1\}, \quad \Sigma_s = \{x : s(x) = 1\}, \quad v_s = \lambda_n(K_s) \in (0, \infty).$$

The finite measure is read as a real number when used as a scalar. Also put $U_p = \{x : p(x) < 1\}$ and $U_q = \{x : q(x) < 1\}$.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B), \mathcal{V}_q(B)$ for its two volume-scaled versions. Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_s(B) = v_s \Pi_n(B) \quad (s \in \{p, q\}) \quad (I = (i_1 < \dots < i_n)).$$

Let $\Delta : \Sigma_p \rightarrow \Sigma_q$ be a bijection preserving chord distances, so

$$q(\Delta(u) - \Delta(v)) = p(u - v) \quad (u, v \in \Sigma_p).$$

Read its radial extension on the reference coordinates:

$$R_\Delta(0) = 0, \quad R_\Delta(x) = p(x)\Delta(x/p(x)) \quad (x \neq 0).$$

For $x \neq 0$, positivity of $p(x)$ makes $x/p(x) \in \Sigma_p$, so this is defined.

Let $A : E \rightarrow F$ be any linear map and put $H = A \circ R_\Delta$. For a function $k : E \rightarrow F$, write $Dk(x)$ for its Fréchet derivative where it exists and for the zero linear map otherwise. Define $G_{p,k}(x) = v_p \Pi_n(Dk(x))$ and $\mathcal{A}_p(k) = v_p^{-1} \int_{K_p} G_{p,k} d\lambda_n$. Then there is one sign $\sigma \in \{1, -1\}$ such that

$$\mathcal{A}_p(A \circ R_\Delta) = \sigma \mathcal{V}_q(A).$$

Assumptions

The dimension is positive: $n = m + 1$. Both p and q are continuous seminorms with the displayed positive norm comparisons. The sphere map is a bijective chord-distance isometry. The map A is arbitrary linear; there is no contraction assumption on it. The output dimension N may be zero.

Conclusion

One scalar sign works for the full vector, including all selected minors. If $N < n$, the vector identity is in the one-element empty-coordinate space; the same scalar construction still applies.

Notes

The derivative and every integral use the fixed sup-norm coordinates and reference Lebesgue measure. The sign is not chosen independently for individual minors.

Proof route

Use signed orientation on open balls, compute linear-composition minors, and cancel the average normalization.

▼ Proof steps

1. **Supply the signed-Jacobian inputs.** Use the same radial map R_Δ and its inverse $R_{\Delta^{-1}}$. The radial norm and inverse identities give

$$q(R_\Delta x) = p(x), \quad R_{\Delta^{-1}}(R_\Delta x) = x, \quad R_\Delta(R_{\Delta^{-1}}y) = y.$$

The global reference-norm bounds are

$$\frac{a_p}{3b_q} \|x - y\|_\infty \leq \|R_\Delta x - R_\Delta y\|_\infty \leq \frac{3b_p}{a_q} \|x - y\|_\infty,$$

$$\|R_{\Delta^{-1}}u - R_{\Delta^{-1}}v\|_\infty \leq \frac{3b_q}{a_p} \|u - v\|_\infty.$$

These follow from the 3-Lipschitz radial estimate in the model norms and the positive comparison bounds. The same identities map U_p onto U_q , with inverse $R_{\Delta^{-1}}$. Both sets are open; U_q is convex and contains zero, hence is preconnected and has positive measure. Also $U_q \subseteq K_q$, so its measure is finite. The exact radial input construction records these inputs.

Write $DR_\Delta(x)$ for the derivative of R_Δ where it exists and for zero otherwise, and set $J_\Delta(x) = \det DR_\Delta(x)$. Apply the signed Jacobian integral theorem with source U_p , target U_q , forward map R_Δ , inverse $R_{\Delta^{-1}}$, and the bounds just displayed. It supplies one sign $\sigma \in \{1, -1\}$ and

$$\int_{U_p} J_\Delta(x) d\lambda_n(x) = \sigma \lambda_n(U_q) = \sigma v_q.$$

For the last equality, the relevant boundary and measure identities are

$$K_q \setminus U_q = \Sigma_q = \partial U_q, \quad \lambda_n(\Sigma_q) = 0, \quad \lambda_n(U_q) = v_q.$$

The null-sphere lemma identifies the sphere with the boundary of the convex open ball and proves that it is null. The open/closed ball-volume identity gives the last equality. No absolute determinant is substituted.

2. **Compute every minor with the same inner determinant.** At a differentiability point of R_Δ ,

$$DH(x) = A DR_\Delta(x), \quad \Pi_n(DH(x))_I = \Pi_n(A)_I J_\Delta(x).$$

Selecting rows commutes with multiplication by the square matrix $DR_\Delta(x)$; determinant multiplicativity then proves the second equality. The exact linear-composition identity gives

$$G_{p,H}(x)_I = v_p \Pi_n(A)_I J_\Delta(x).$$

The global Lipschitz bound on R_Δ gives differentiability almost everywhere. The linear map A is Lipschitz, so H is Lipschitz. Compact-set integrability of derivative generators applied to H on K_p makes the vector $G_{p,H}$ integrable. Apply the same result to $R_\Delta : E \rightarrow E$ on K_p , and let $I_0 = (1, \dots, n)$ be the all-row index. Its determinant coordinate gives

$$G_{p,R_\Delta}(x)_{I_0} = v_p J_\Delta(x), \quad J_\Delta = v_p^{-1} (G_{p,R_\Delta})_{I_0} \in L^1(K_p, \lambda_n).$$

Coordinate projection and multiplication by the finite constant v_p^{-1} preserve integrability. Thus both the scalar integral and the vector coordinates used below are justified.

3. **Cancel the normalization and insert the one sign.** For every $I \in \mathcal{I}_{n,N}$,

$$\begin{aligned} (\mathcal{A}_p(H))_I &= v_p^{-1} \int_{K_p} v_p \Pi_n(A)_I J_\Delta d\lambda_n \\ &= \Pi_n(A)_I \int_{K_p} J_\Delta d\lambda_n \\ &= \Pi_n(A)_I \int_{U_p} J_\Delta d\lambda_n \\ &= \sigma v_q \Pi_n(A)_I. \end{aligned}$$

The first two equalities use integrability and $v_p > 0$. The third discards exactly $K_p \setminus U_p = \Sigma_p$, whose measure is zero by The null-sphere lemma. The coordinate radial-average theorem records those first three lines; the last substitutes the signed integral from Step 1. The sign was chosen before any row index, so it is common to every coordinate. Reassembling the vector gives $\mathcal{A}_p(H) = \sigma \mathcal{V}_q(A)$.

Main citations

- Exact declaration and its source — `MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial`
- The exact radial input construction — `MathlibAnnex.Plucker.Internal.radialOpenData`
- the signed Jacobian integral theorem — `MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume`
- The exact linear-composition identity — `MathlibAnnex.Plucker.derivativeGenerator_comp_linear_apply`
- Compact-set integrability of derivative generators — `MathlibAnnex.Plucker.integrableOn_derivativeGenerator_compact`

- The coordinate radial-average theorem — `MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial_apply`
- The null-sphere lemma — `MathlibAnnex.Plucker.Internal.sphereSet_null`
- The open/closed ball-volume identity — `MathlibAnnex.Plucker.modelOpenBall_volume_eq_ballVolume`
- Set-average definition — `MathlibAnnex.Plucker.derivativeAverage`

Lean source signature (exact)

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/-- The full radial average is exactly one orientation sign times the target
generator. No contraction hypothesis on the linear map is introduced. -/
theorem derivativeAverage_comp_linear_radial {m N : ℕ}
  {MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)}
  (Δ : sphere (0 : Space MX) 1 ≈i sphere (0 : Space MY) 1)
  (A : (Fin (m + 1) → ℝ) →L[ℝ] (Fin N → ℝ)) :
  ∃ ε : ℝ, (ε = 1 ∨ ε = -1) ∧
    derivativeAverage MX (fun x => A (show Fin (m + 1) → ℝ from
      radialExtension (X := Space MX) (Y := Space MY) Δ (show Space MX from x))) =
      ε • Matrix.ballVolumeScaledMaximalMinors MY A

```

The inputs specify two norms, a sphere isometry and an arbitrary linear map. The conclusion is an existence statement about a single sign.

In the source	Mathematical meaning
<code>{m N : ℕ}; Fin (m + 1) → ℝ; Fin N → ℝ</code>	$m, N \geq 0, n = m + 1 > 0, E = \mathbb{R}^n, F = \mathbb{R}^N$. Coordinate vectors have the sup norm; braces mark parameters Lean can infer.
<code>MX; MY</code>	The continuous seminorms $p = \text{MX} \cdot p$ and $q = \text{MY} \cdot p$, with their positive lower and upper comparison constants stated in this Card.
<code>Space MX; Space MY</code>	The same vectors of E , now with norms p and q , respectively.
<code>sphere (0 : Space MX) 1; sphere (0 : Space MY) 1</code>	$\Sigma_p = \{x : p(x) = 1\}$ and $\Sigma_q = \{x : q(x) = 1\}$. The arguments are center 0 and radius 1.
<code>Δ : ... ≈_i ...</code>	A bijective isometry $\Delta : \Sigma_p \rightarrow \Sigma_q$; $q(\Delta u - \Delta v) = p(u - v)$ for all $u, v \in \Sigma_p$. The symbol $\approx_i$ in the code asserts both bijectivity and distance preservation.
<code>A : ... →L[ℝ] ...</code>	A continuous real-linear map $A : E \rightarrow F$; no contraction inequality is assumed.
<code>show Space MX from x</code>	The same coordinate vector x , regarded as having norm p .

In the source	Mathematical meaning
radialExtension (X := Space MX) (Y := Space MY) Δ	The map R_Δ , with $R_\Delta(0) = 0$ and $R_\Delta(x) = p(x)\Delta(x/p(x))$ for $x \neq 0$. The named arguments specify the source norm p and target norm q .
show Fin (m + 1) $\rightarrow$ $\mathbb{R}$ from ...	Read the resulting vector in the original sup-norm coordinates. No additional mathematical map is applied.
fun x => A (...)	The entire function $H : E \rightarrow F$, $H(x) = A(R_\Delta(x))$; fun x => means $x \mapsto$.
derivativeAverage MX (...)	$\mathcal{A}_p(H) = v_p^{-1} \int_{K_p} v_p \Pi_n(DH(x)) d\lambda_n(x)$, where $K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda_n(K_p)$. The derivative is read in the reference coordinates, with zero at nondifferentiability points.
Matrix.ballVolumeScaledMaximalMinors MY A	$\mathcal{V}_q(A) = v_q \Pi_n(A)$, where $v_q = \lambda_n(\{x : q(x) \leq 1\})$ and $\Pi_n(A)_I = \det((Ae_j)_{i_a})_{a,j=1}^n$.

In the source	Mathematical meaning
$\exists \varepsilon : \mathbb{R}, (\varepsilon = 1 \vee \varepsilon = -1) \wedge \dots$	There exists a real $\sigma = \varepsilon$ with $\sigma = 1$ or $\sigma = -1$, and the following vector equality holds for that same sign. Here $\vee$ means “or” and $\wedge$ means “and”.
$\dots = \varepsilon \bullet \text{Matrix.ballVolumeScaledMaximalMinors MY A}$	$\mathcal{A}_p(H) = \sigma \mathcal{V}_q(A)$. The dot $\bullet$ multiplies every minor coordinate by the one scalar σ .

Exact definitions used in this reading: Set-average definition. The mathematical proof keeps R_Δ and $R_{\Delta^{-1}}$ for these same radial maps. Its signed integrand $J_\Delta(x) = \det DR_\Delta(x)$ is distinct from the absolute integrand $j_\Delta(x) = |\det DR_\Delta(x)|$ used in the absolute-area Card. The single output sign is shared by all minor coordinates.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 16–19:

```
noncomputable section
open Set Metric Function MeasureTheory
open scoped NNReal ENNReal BigOperators
namespace MathlibAnnex.Plucker
```

▼ Exact content identity

Declaration: MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial

Accepted content SHA-256: 0e7ea02ecc81f88e0f8ec2c926430d384d7e72de01feb5d2563920f636facc9b

Accepted source guide SHA-256: 9bd09b5ac6b436fd4a74c33f8f6759541be90735a9ec39b3c044e1dd14fb8b64

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73